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Safety assessment of MSR concept using INPRO methodology

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ABSTRACT

The share of energy from nuclear source is constantly increasing globally. With the advancement in every walk of life, be it social; economic or technological; there is one common factor which is energy. To meet these energy needs, nuclear power has to contribute its share in a sustainable way. In order to assess the sustainability of any nuclear energy system (NES), IAEA's International Project on Innovative Nuclear Power Reactor and Fuel Cycles (INPRO) has developed a methodology known INPRO methodology. It covers seven different areas including economics, infrastructure, safety, waste management, environment, physical protection and proliferation resistance. In the present paper, safety of molten salt reactor (MSR) concept has been assessed using INPRO methodology. MSR is one of the six designs being pursued by GEN IV Forum. All the basic principles, user requirements, criteria in the area of safety of INPRO methodology have been assessed based on the data available in open literature. It is observed that the MSR technology ticks most of the boxes in terms of safety. However, some gaps have also been identified which need further R&D for this concept.

1. Introduction

With the increase in energy demand, the share of energy from nuclear sources is increasing day by day. Today we see applications of nuclear energy in various fields such as agriculture, medicine, power sector, desalination, food irradiation and nuclear space reactors etc. In the power sector, conventional sources of energy like oil and gas are neither sustainable nor renewable, while the nuclear energy has the potential to serve as both. Nuclear energy provides reliable and base-load electricity. Energy from nuclear source is available for long hours and continuously. It is a green energy, causing no danger to our environment in terms of emission of the greenhouse gases (GHGs). In comparison to other energy sources, fuel consumption is very low in nuclear energy systems; hence, it can provide immense power in a relatively small volume compared to other energy sources. At present, 12% of the world's electricity is generated by nuclear and this much electricity is being produced from 448 nuclear reactors globally (International Atomic Energy Agency, 2017).

With the word nuclear comes the negative sentiment of general public as nuclear industry has suffered its share of unfortunate events in the past. Accidents like Three Mile Island (TMI),Chernobyl and Fukushima raise question against the safety of nuclear reactors. As discussed, to fulfill the higher energy demand in various walks of life requires, the use of nuclear is essential. However, in order to deploy nuclear reactors successfully around the globe to meet high energy demands, public sentiment towards nuclear has to be changed. In this regard, GEN IV International Forum (GIF) has identified eight different technology goals in the areas of sustainability; safety; economics; and physical protection and proliferation resistance (Levy and TodreasRalph Bennett, 2001). These goals have been defined keeping in view the social, economic and environmental requirements of the 21st century. GIF has selected six different reactor concepts which would be safer, cleaner, cost-effective, and more resistant against proliferation and terrorist attacks compared to existing reactors. These reactors are designed with lower Core Damage Frequency (CDF) and lower Large Early Release Frequency (LERF). These reactors are judged by deterministic and probabilistic approaches at every step of their development, ultimately ensuing in a safe and sustainable reactor.

Since nuclear power is considered to be a sustainable option and allows the nations to achieve the goals of sustainable development, therefore it is important to check the sustainability aspect of any nuclear energy system (NES) including innovative nuclear energy system (INS) used for generating electricity. Sustainable development is defined as: *Development that meets the needs of the present without compromising the ability of future generations to meet their own needs* (International Atomic Energy Agency, 2008). Sustainable development is linked with social, environment, and economic dimension. In year 2000, the International Atomic Energy Agency (IAEA) launched the

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International Project on Innovative Nuclear Power Reactor and Fuel Cycles (INPRO). INPRO has developed a methodology known as"INPRO Methodology" which assesses the sustainability of any NES. The idea of developing this methodology was to evaluate NES in the light of environmental impact, safety, infrastructure, economics, waste management, and proliferation resistance. In this connection, different requirements and indicators were developed for each area. These indicators are assessed against some acceptance limit and as a result NES is declared sustainable for that particular area.

In the present study, safety of molten salt reactor (MSR) concept has been assessed using requirements and indicators present in the area of safety of INPRO methodology. All the subsequent indicators designed in the area of safety revolve around the basic safety requirement, which is to protect people and environment from harmful effects of ionizing radiations (safety fundamentals). And the choice of molten salt reactor (MSR) concept has been motivated by the previous tests performed on molten salt reactors at Oak Ridge National Laboratory (ORNL) and a current increase in interest in molten salt reactors. The assessment is performed based on the data available in open literature.

2. Molten salt reactor (MSR)

Lots of work has been done on MSRs in the past. The programs such as Aircraft Reactor Experiment (ARE) from 1953 to 1954 and Molten Salt Reactor Experiment (MSRE) from 1960 to 1969 are good examples of developmental work related to MSRs. The MSRE was in operation for 4.5 years before it was shut down. In ARE, NaF-ZrF4-UF4 (53-41-6 mol %) was used as fuel salt; BeO as moderator; liquid sodium as secondary coolant; and Inconel-600 was used for structures and piping. In MSRE, LiF-BeF2-ZrF4-UF4 (65-29-5-1 mol%) was used as fuel salt while 2LiF-BeF2 was used as secondary salt. Metallic parts in contact with salt were made of Hastelloy-N. Uranium was removed from salt through fluorination. In 2007, ORNL came up with an MSR design that would use LiF-BeF2-ThF4-UF4 (72-16-12-0.4) as a fuel (Serp et al., 2014; World Nuclear Association, 2016).

The lucrative advantages in the use of MSRs have captured the attention of the world once again. MSRs are inherently exempted from severe accidents like core-melt. Operating pressure in MSRs is significantly lower than in conventional light water reactor (LWR) reactors. Ultimate passivity in shutting down an MSR and lower excess reactivity values are other advantageous features of MSRs over LWRs (Serp et al., 2014). Due to these advantages, this reactor concept is also present in the six designs which are being actively pursued by GIF for future commercialization.

Various organizations in many countries around the world are working on different designs of MSRs. For example, France and Switzerland are working on Th-U Molten Salt Fast Reactor (MSFR) (Merle-Lucotte et al., 2013). Russian Federation is working on Molten Salt Actinide Recycler and Transmuter, both with and without the use of Th-U. China, South Korea and Japan are working on Th-U Thorium Molten Salt Reactor (TMSR). All these concepts involve fuel salt i.e. liquid fuel. A European initiative on Safety Assessment of Molten Salt Fast Reactor (SAMOFAR) is dedicated to experimental proof of concept of MSFRs. With solid fuel, both USA and China are carrying out R&D for FHR (Fluoride-salt-cooled High-temperature Reactor) (Ignatiev, 2016a). A short overview of some of the concepts which use molten salt technology is given below in Table 1. This data has been taken from IAEA's database on Advanced Reactor Information System (ARIS).

The concept of Molten Salt Actinide Recycler and Transmuter (MOSART) is advantageous as it can use spent fuel of current LWRs. The transuranics of UOX from a PWR spent fuel after cooling can be used as fuel for MOSART. The fuel salt in MOSART is LiF-BeF2-NaF (15-27-58 mol %) (Ignatiev et al., 2003). It is fueled by the trifluorides of actinides. The metal surfaces in contact are made of an alloy, which is a modified form of Hastelloy N. It has a homogenous core with a fast neutron spectrum. This design will help in reducing the toxic effects of long-lived radioactive isotopes by putting them into use for producing electricity (Ignatiev, 2016b).

3. Overview of INPRO methodology

In year 2000, an international project on innovative nuclear reactors and fuel cycles (INPRO) was launched by the IAEA. The main objectives of INPRO are (International Atomic Energy Agency, 2008):

- *To help ensure that nuclear energy is available to contribute in a sustainable manner to the energy needs of the* 21st *century;*
- *To bring together technology holders and users to consider jointly national and international actions required for achieving desired innovations in nuclear reactors and fuel cycles.*

INPRO developed methodology in 2006 which serves as a tool to assess innovative nuclear energy system (INS). The purpose behind developing methodology was to address some ofthe concerns associated with nuclear power related to economics, safety, waste etc. In this regard, INPRO selected seven (07) different areas to address all those concerns. These areas include economics, infrastructure, waste management, proliferation resistance, physical protection, environment, and safety. In all of these areas, INPRO came up with different manuals containing basic principle, user requirements, criteria, specific indicators and acceptance limits that help in assessing the sustainability of any INS. Assessment using INPRO methodology can be performed for all innovative reactors as well as evolutionary reactors. It can also be used for nuclear fuel cycle facilities. The link between INPRO areas and general concept of sustainability with its dimensions is given in Fig. 1 while the architecture of INPRO methodology is shown in Fig. 2.

In INPRO methodology, basic principle defines the goals for developing a NES. On the other hand user requirements ask the designers; operators; industry and/or a state that what needs to be done to meet the goals defined in basic principle. Criteria which consist of indicators and assessment limit allow the assessor to check if a particular user requirement is met or not (International Atomic Energy Agency, 2008).

4. Assessment of safety of MSR concept

The subject area of INPRO methodology that we are concerned within this paper is the area of safety. INPRO basic principles and user requirements in the area of safety are designed on the basis of IAEA fundamental safety objectives and principles, and indeed a great tool for assessing the safety of innovative nuclear reactor designs. MSR concept has been assessed against all the basic principles, user requirements, and criteria mentioned in the safety manual of INPRO

Table 1
Overview of some molten salt reactor concepts.

Reactor	IMSR-400	LFTR	SmAHTR	MSFR	MSR-FUJI
Designer country	Canada	USA	USA	France	Japan
Fuel type	UF4 in diluent fluorides	Th-U cycle	TRISO	Actinide fluorides of Th & U-233	Th-U molten Salt
Neutron spectrum	Thermal	Thermal	Thermal	Fast	Thermal
Moderator	Graphite	Graphite	Carbon	No solid moderator	Graphite

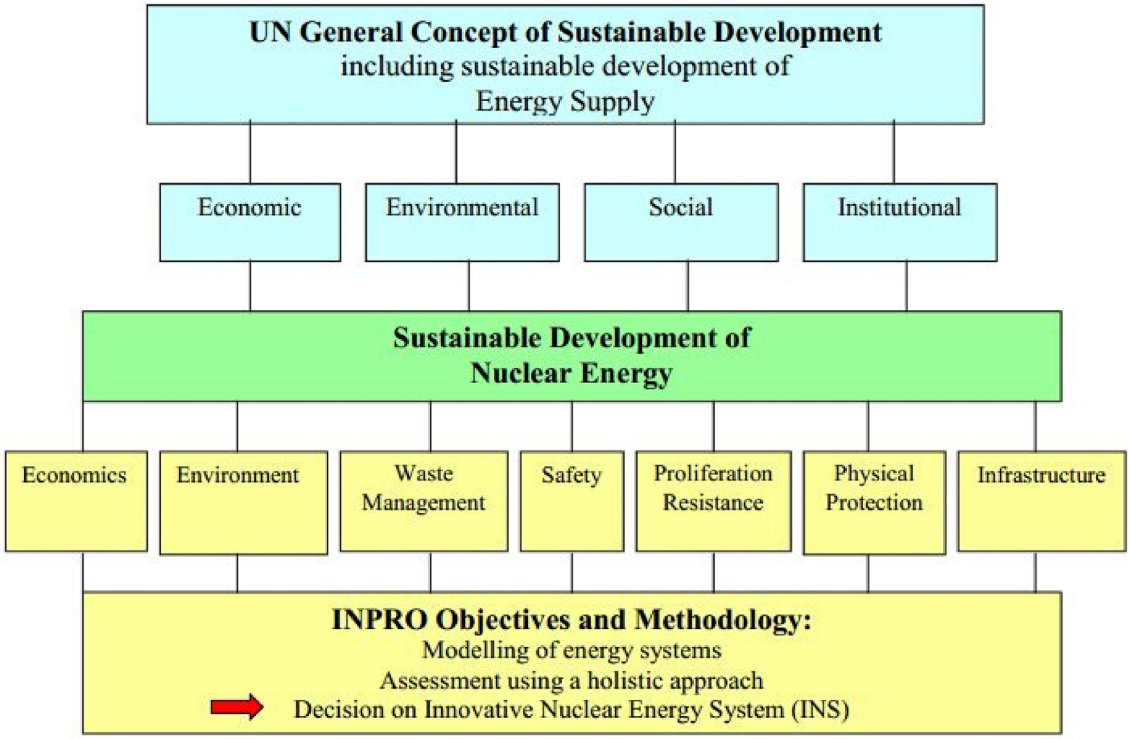

Fig. 1. Interrelationship between UN concept of sustainability and INPRO (Nuclear Power Technology Developing Section, 2008, International Atomic Energy Agency, 2008).

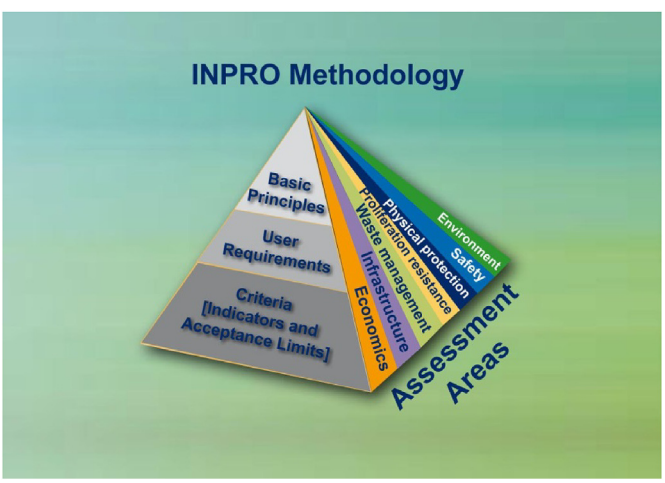

Fig. 2. Architecture of INPRO methodology (International Atomic Energy Agency, 2009).

methodology (IAEA International Atomic Energy Agency, 2008). At present, all MSR designs are in conceptual design stage; therefore, instead of focusing on a single design, INPRO requirements have been applied on overall MSR concept.

4.1. Results of assessment

4.1.1. Basic principle 1 – defense in depth

UR 1.1 – robustness, CR 1.1.1 -robustness. Molten salt reactors have significant margins of design. The margin is the difference between the operating and design values. The higher margin means better robustness. These are helped in case of molten salt reactors by the strong negative temperature and void coefficients (LeBlanc, 2010). Negative temperature coefficient is strong mainly due to the fertile material (usually thorium) absorbing more neutrons when overheated owing to the "doppler effect". The recirculation of fuel salt allows for the neutron precursors to move away from active region. Fuel being in liquid state allows for greater thermal expansion in comparison to LWRs. This feature helps in insertion of negative reactivity in the core. Besides fission product poisoning, load following ability is hindered by pellet-cladding interactions (PCI) in LWRs. The pellet undergoes quicker contractions and expansions with changes in power causing the pellet to push strongly towards cladding material, which may lead to release of corrosive fission gases. However in case of MSRs, load following is more attractive as the fuel is in liquid state (Lokhov, 2011). Considering the thermal fuel design, MSRs have a more uniform burnup of fuel not allowing any excessive centerline temperatures, as is the case in current LWRs. At every moment, a different part of the fuel experiences maximum temperature because of the flow. The liquid state also eliminates the use of special cladding for each fuel assembly. This eliminates an extra resistance to heat transfer from the fuel. Unlike solid fuel reactors, there is no risk of any mechanical failure of the fuel (LeBlanc, 2010; Elsheikh, 2013).

In the current design of fuel assemblies, fission gases are trapped inside the assembly and provisions are made for this purpose as well. This is taken care of by the online processing ability that MSRs can provide. The predecessor at ORNL used processes such as 'fluorination' and 'vacuum distillation' to remove volatile fission products, such as rare earths and fission gases from the fuel (Carter and Scott and, 1966). For example, in the two-fluid design at ORNL, before the criticality of MSRE, the U-233 formed in the blanket salt was transferred to the fuel by 'fluorination'. In this process fluorine gas is bubbled through the blanket salt. This converts UF4 into UF6 and then this UF6 is converted back to UF4 before transferring it back to the fuel salt. The ability to have such processes going on while the reactor is operational removes any use of the fuel handling equipment, helping in improving the robustness of design (Serp et al., 2014). The quality of salts needed to begin operation of an MSR has already been achieved, but for improved quality further research is required. China is producing mass quantities

of salt for such purposes on a regular basis now. For further development of materials required for fluoride salt cooled high temperature reactor (FHRs), Australia has collaborated with China in 2016 (Ignatiev, 2016b).

CR 1.1.2 – operation. Margins in terms of operation of MSRs are also very promising. The operating temperatures of 970 K and boiling temperatures of fuel salt around 1670 K allows room for transient control (Elsheikh, 2013). Additionally, if online processing is available, the fuel composition can be varied anytime in any proportions. This allows for lower excess reactivity in the core and lower control rod worth, reducing the reactivity shift by control rods (Doligez et al., 2014). In terms of availability; excellent load following capability of MSRs and no need for refueling outages keeps the plant available. This feature also helps in designing plants where the initially loaded fuel would suffice for entire life of the core, not requiring any refueling outages (RFO) as required in LWRs. As compared to current PWRs, a similar transient would have lower impact on MSRs owing to stronger negative coefficients (Lelek, 2001).

CR 1.1.3 – inspection. MSRs have very low operating pressure; in some designs even near atmospheric pressure. With operating pressures falling under 5 bars, the conditions for inspection and monitoring are not as harsh. The temperatures inside are significantly higher and should be kept in safe limit. However, there is a difficulty in inspection due to higher temperatures involved. But because of the experience from other industries dealing with high temperatures, the instrumentation framework is more established and it has shown to work well with molten salt reactors (Holcomb et al., 2018). Appropriate and complete inspection program for MSRs includes knowledge of materials used, NDT results, damage mechanisms, location with high stresses, and risk informed studies. Such detailed information is not yet available. However, due to less severe pressure conditions and no need for refueling, it can be envisaged that the inspection program will serve the purpose quite well (Boyer, 2016).

CR 1.1.4 – failures and disturbances. The concept of MSRs in itself eliminates most of the causes for disturbances. Some of them are (Elsheikh, 2013):

- Operating pressures of less than 5 bars reduces the likelihood of over pressurization, thus, lowering the probability of accidents such as system/component destruction or salt leakage or fuel release, which complies well with INPRO acceptance limits.
- Containment integrity would not be threatened due to excessive pressure.
- Fission product gases are continuously removed; lowering the risk of release to environment in abnormal cases.
- Lower excess reactivity in comparison to LWRs which reduces the reactivity shifts in case of transients.

4.1.2. UR 1.2 – detection and interception

CR 1.2.1 – I&C and inherent characteristics. The main function of instrumentation and control is to detect anticipated operational occurrences and return the plant to normal operating conditions. The inherent characteristics of molten salt reactors make it very easy for I&C to perform this function. The combination of active processing and passive shut down system helps in maintaining the normal operating conditions. Moreover, lower pressures help in vibration monitoring as well as loose part monitoring. Instrumentation is more stressed at high temperature but the detailed framework is being worked out with plenty of success (Holcomb et al., 2018). With the current conceptual designs, we can have a remotely monitored plant with limiting inherent characteristics i.e.the one in which the operator would not need to go near the core area in the entire life of the plant.

CR 1.2.2 & CR 1.2.3 – grace period and inertia. Grace period of MSRs is significantly greater than LWRs. This is due to the fact that in any abnormal case, the reactor can be shutdown passively. One of the means is 'Freeze plug'. INPRO acceptance limit for grace period is 30 min, but with MSRs (e.g. ThorCon) a grace period of 72 h is achievable (Nuclear Power Technology, 2016). In case of transients, the low power density of MSRs ensures that operators have sufficiently large grace period and inertia (Forsberg et al., 2003). Also, the strong negative coefficients and inert nature of the salts help in increasing inertia of the reactor.

4.1.3. UR 1.3 – design basis accidents

CR 1.3.1 – frequency of DBAs. MSRs, by design, reduce the frequency of some design basis accidents (DBAs). Some encouraging features of the molten salt reactor concept are:

- Considering the lower operating pressures, causes for all kinds of rupture are eliminated in MSRs.
- The fuel is already in molten state; hence, no danger of having fuel distorting into any shape, way or form.
- The control rod ejection and other related accidents are avoided by lowering the control rod worth (reactivity induced accidents) (Elsheikh, 2013). Other transients like loss of power and failure of pumps that may introduce positive reactivity into the system have also been studied for molten salt reactors. The frequency of release due to loss of offsite power (LOOP) is lower for MSRs in comparison to LWRs. These accidents are discussed further under CR1.6.1. The insertion of positive reactivity due to overcooling of salt is also nullified by the large negative feedback coefficients of molten salt reactors (Cao et al., 2017).
- With the facility of online processing, any fuel composition can be made any time –allowing greater control.
- With no water and hydrogen-producing-cladding, the chances of hydrogen explosion are highly reduced (Ottewltte, 1992).
- In case any thing goes severely wrong, the reactor can be shut down without the intervention of operator. In fact, there is nothing that the operator can do to stop the reactor from shutting down then (Elsheikh, 2013).

One of the potential problems with MSRs is production of tritium, which is estimated at 2400 Ci/day for 1000 MWe reactor. Tritium is mainly produced from Li^6(σ = 940b)– a strong neutron absorber. If lithium is to be used in MSRs, the focus will be to use Li^7(σ = 20 mb) as much as possible to avoid undesired tritium production (Schmutz et al., 2012). Another possible cause of tritium production might be beryllium, which in some designs like TAP (Trans Atomic Power Reactor) is chosen to be eliminated (Diamond, 2018). However, this is one area where further R&D is required as pointed out in CR 4.2.1.

CR 1.3.2 – grace period. Already discussed under CR 1.2.2 & CR 1.2.3.

CR 1.3.3 – safety features. As per INPRO requirements, an INS should have enhanced safety features involving a combination of active and passive safety systems. In this area, MSRs are the most attractive option among all of the GEN IV concepts. Major safety features of MSR designs are as follows (Elsheikh, 2013):

- The prompt neutron life time is greater in MSRs, which also increases the safety in terms of controlling neutron flux.
- In case of Reactivity Insertion Accidents (RIA) (e.g. control rod ejection/withdrawal accident) the lower control rod worth means smaller transients as compared to LWRs.
- If all the pumps used for primary salt circulation trip, the temperature inside reactor vessel will start to increase. This has the same effect as that of reactivity insertion. The transients caused are smaller in case of MSRs because of the lower worth of control

elements and the capability of keeping the reactivity low while the reactor is running.

- Fuel salt leak accident – for reasons other than pressure, the leaked salt is caught by a "catch-pan", which directs the hot salt to non-critical drain tanks.
- Freeze plug/fuse valve: It is normally in solid form blocking the passage to drain tank, but when the temperatures increase it melts and pours the salt down to a drain tank. As soon as the salt is in the drain tank, fission stops because of no moderator there. Freeze plug works on the principle of gravity, making it most reliable in terms of passive safety.
- Gaseous fission products are continuously removed from the system; hence, it results into a minimum danger of radioactivity release in accidental conditions (Doligez et al., 2014).
- The molten salt is chemically inert. It does not react with water and is not flammable (Minck, 2013). Furthermore, corrosion problems (as experienced in ARE) can be minimized by proper chemistry control.
- The salt seeking fission products like Sr-90 and Cs-137 are bound to the salt. In case of spillage, they go down to the drain tanks with the salt (Ottewltte, 1992).
- The above mentioned safety features eliminate the need for diverse and redundant safety like ECCS. This also simplifies the design.
- Containment is equipped with heaters to avoid freezing accidents of molten salts.

CR 1.3.4 – barriers against release. According to this criterion, at least one barrier should be there in case of severe accident conditions. This is easily fulfilled because of lower pressures and lower source term in molten salt reactors. An idea of having barriers in MSR against release is different from the current reactor. The concept of core meltdown does not apply to MSRs. Although there are physical barriers of reactor vessel and containment, one can see the need of new definitions for barriers in case of molten salt reactors (MoirTeller, 2018).

CR 1.3.5 & CR 1.3.6 – controlled state and sub-criticality. A controlled state is a state at which the safety features of the reactor are able to compensate for the accidental conditions. In molten salt reactors, for the most severe conditions, there is a freeze plug or fuse valve to safely bring the reactor to shutdown state with no need of operator's intervention. Even if there is no power at that moment, this passive system can still shut the reactor down. Once the molten salt has been poured into the drain tanks, the passive heat removal system around it will take care of the decay heat and the geometry configuration will keep the salt sub-critical. Acceptance limit as per INPRO requirement is that the shutdown reactivity margin has to be less than 1%$\Delta k/k$. This condition applies only in worst accidental cases (IAEA International Atomic Energy Agency, 2008).

Excess reactivity in the core is needed for smooth operation of the reactor over the entire cycle length. But it is important to have sufficient shutdown margin to make the reactor sub-critical if needed. For molten salt reactors, the excess reactivity is kept to a minimum value which helps in achieving safety in accidental conditions. On top of that the drain tanks can be designed so that the fuel salt becomes sub-critical as soon as it is collected. This has been verified by Okun and Utah in 2014 using the specifications of MSRE at ORNL using MONK version 9A Monte Carlo Program. The effective multiplication factor in the non-critical drain tanks came out to be 0.9076 (sub-critical) without the usage of control rods which agrees well with the INPRO requirements (Okon and Uwah, 2014).

4.1.4. UR 1.4 – release into containment

CR 1.4.1 – major release into the containment. In case of molten salt reactors, the probability of major release into containment can significantly be lowered because of low pressure and smaller source term (Elsheikh, 2013).

CR 1.4.2 – processes. Source term is the driving force for dispersion and it is also a measure of the heat or energy that would damage the reactor in the event of heat removal failure. With liquid fueled reactors, there is an option of online processing that can continuously remove the insoluble fission products such as Xenon and Krypton while most of the other fission products are soluble in the salt. Thus, it helps in reducing the probability of release (Gat and Dodds, 2008).

CR 1.4.3 – accident management. In case of accidental conditions, the core can be brought to controlled state at shutdown using the drain tanks and then cooled down. As a result, it would not allow any activity to be leaked out (Okon and Uwah, 2014). As per INPRO requirement of technology availability, the predecessors of MSRs used this technique for shutting down the experimental MSR at ORNL (Elsheikh, 2013).

4.1.5. UR 1.5 – release into environment

CR 1.5.1, CR 1.5.2 & CR 1.5.3 – release into environment. The safety concept of MSRs practically eliminates the possibility of release (ARIS database, IAEA). Even after the life of the reactor, the fuel is relocated to the specialized facility not allowing any significant activity to be leaked out. There is still a potential for the fission products to leak out from places other than the primary loop but designing against such release is easier because of lower driving forces. Similar to the other liquid fuel reactors, the evacuation zone for an MSR is greatly reduced because of the elimination of the major factors driving the release of radioactivity. No design has yet been completely analyzed but the minimal risk in terms of release is what makes the MSRs an attractive option.

4.1.6. UR 1.6 – independence of DID levels

CR 1.6.1 – independence of DID levels. No PSA or hazard analysis has yet been done on MSRs thoroughly. But the work carried out so far has shown outstanding prospects. One of the purposes of PSA is to demonstrate reliability of inherent characteristics and passive safety systems. Loss of offsite power (LOOP) is one possible accident to any type of reactor. Mudan et al. (Mei et al., 2013) performed PSA of LOOP in case of a molten salt reactor. The research concluded that the release of molten salt radioactive material to the core due to LOOP is 2×10^{-12}/reactor-year, which is lower than conventional LWRs. The study further concluded that one of the major causes of release to the core is cavity (reactor) cooling function failure. To study the cooling capability under accidental conditions, the investigation of natural circulation capability of molten salt reactor was performed under the umbrella of SAMOFAR initiative, which assumed the failure of pumps – another possible accident. It was concluded that the reactor is capable of cooling itself down after the pump failure. The flow is disturbed for the first 5 s but then established again due to natural circulation. The fuel does not overheat in this case, not even exceeding the maximum temperature of the fluid (Nijen, 2018). A deeper analysis of passive fuel drainage system of MSR was performed by Oliver van den Bergh (2016). As Brovchenko et al. (2013) suggested that the fuel must be drained before 8 min after a transient has occurred. Oliver van den Bergh (2016) showed that the drainage time for MSFR is between 8 s and 121 s, well below 8 min and that there is no blockage of drainage pipes due to salt crystallization (Oliver van den Bergh, 2016).

PSAs are normally supplemented by qualitative techniques and the techniques indicate more independence of DID levels in MSRs. Residual heat removal system that run passively, the drain tanks that are filled passively due to gravity, the online-processing allowing for low excess reactivity and quite less human intervention are some of the features of MSRs. Introducing passive safety and reducing the human efforts required generally help in providing independent DID levels.

4.1.7. UR 1.7 – human machine interface

CR 1.7.1 – human factors. Human intervention in terms of attaining a controlled state is almost eliminated (Okon and Uwah, 2014). In terms

of plant life, no specific data available, as no design has yet matured.

CR 1.7.2 – human response model. The use of artificial intelligence is a very important factor in the response model of any reactor and with MSRs, this is quite achievable. Human intervention in terms of fuel loading and reloading responsibilities has been eliminated (Nuclear Power Technology, 2016). With some conceptual designs capable of keeping fuel salt in tanks for 72 days, there is lesser dependence on operators and short-term accident management.

4.2. Basic principle 2 – inherent safety

4.2.1. UR 2.1 – minimization of hazards

CR 2.1.1 & CR 2.1.2 & CR 2.1.3 – hazards, frequencies of AOO and consequences. Inherently safe characteristics of molten salt reactors eliminate most of the hazards associated with conventional LWRs and lower the frequencies of other anticipated operational occurrences. These inherently limiting characteristics lower the risk and the consequences of accidental conditions (Elsheikh, 2013).

4.3. CR 2.1.4– confidence in innovation

In comparison to other Gen IV concepts, MSRs had been operated to perform favorably. The initial experimental reactors at ORNL including ARE and MSRE confirmed the operability of MSRs (World Nuclear Association, 2016). At present, most of the design of MSR has been completed. The major hurdles in implementation of molten salt technology are the licensing and material compatibility. The framework of licensing new reactors is based on solid fuel reactors. This makes it important to introduce some new regulations for liquid fuel (Elsheikh, 2013). Further research is being carried out on MSRs in countries like USA, Russia, China, Japan and France.

4.4. Basic principe 3 – risk of radiation

4.4.1. CR 3.1.1 & CR 3.2.1 – occupational dose and dose to public

In terms of doses, current reactor designs have already achieved the limits specified by the regulatory authorities. INPRO does not require a further decrease for a reactor to be declared safe. The only restriction is not to exceed these dose limits (IAEA International Atomic Energy Agency, 2008). The conservative approach would imply similar doses from molten salt reactors as we have from current reactors. But realistically, the automation and introduction of passive safety features in GenIV concepts will have even lower doses to workers and public. Some conceptual designs like ThorCon and IMSR-400 have reported occupational doses of less than 1 person-Sv/year (≈10 mSv/year) (Nuclear Power Technology, 2016).

4.5. Basic principle 4 – RD&D

4.5.1. UR 4.1 – safety basis

CR 4.1.1 – is safety concept defined?. The answer is no. The process of safety concept is iterative. At current stage of conceptual design; preliminary codes, analytical models and supporting data must be developed. This has been done for MSRs. The complete uncertainty-free models should be present before commercialization. In case of MSRs, the uncertainties are only due to the lack of proof of concept, which is currently being followed vigorously. Hence considering the current stage of MSRs, it agrees with the INPRO requirements. However, for commercialization, MSRs do not satisfy the INPRO criteria.

CR 4.1.2 – are safety issues addressed?. Safety issues have been identified and are being addressed.

4.5.2. UR 4.2 – RD&D for understanding

CR 4.2.1 – RD&D. In the past 30 years, most of the technological gaps have been recognized and are being worked on e.g. (Forsberg, 2006):

- Freeze protection
- Tritium control
- Chemical reactions
- Noble metal plate-out
- High level waste form
- High temperature compatible materials

Workable MSRs were developed in 1950s and 1960s with significant operational challenges. In the last 40 years, commercial viability of MSRs has improved significantly in comparison to other reactor concepts. China and USA have signed CRADA for solid fuel MSRs (Ignatiev, 2016a). CAS is working on TMSR, with a 2 MW MSR to be ready by 2022 (Yu and Xiaohan, 2016). Europe is focused on liquid fuel with no solid moderator. Russia is working on MOSART.

CR 4.2.2 – computer codes. No data on specific codes is available. But for some systems the research is going on. For example, analysis of passive residual heat removal system has been done for both steady and unsteady cases (Wu et al., 2016; Wang et al., 2013). In another study, a CFD analysis was performed by Jayaraj et al., showing that power transients both cases (i.e. step up and step down) do no destabilize the system (Yallappa Kudariyawar and Kumar Sriva, 2016).

CR 4.2.3 – scaling. The test performed for any reactor start at a smaller level. Then to extrapolate those results scaling is performed. In case of MSRs, full scale facilities of some small reactors like liquid fluoride thorium reactor (LFTR), ThorCon and IMSR-400 are in vision but all are at the conceptual design phase with considerable R&D going on. FUJI (International Thorium Molten Salt Forum, Japan) built three experimental MSRs but the detailed design has not been worked on. No full-scale test results are available so far.

4.5.3. UR 4.3 – pilot plant

CR 4.3.1 & CR 4.3.2 – novelty and pilot facility. Novelty is described in terms of high and low degree. High degree of novelty means innovative idea. Low degree of novelty means that results from other facilities can be extrapolated to the current design. In case of molten salt reactors, high degree of novelty has been introduced. This requires that a separate pilot facility be built for MSRs, as per INPRO requirements. No such progress has been demonstrated yet. Many countries, for example China etc., are working on the pilot facility. Some designs even use MSRE of ORNL as their pilot facility. However, current results are not sufficient to be extrapolated (Nuclear Power Technology, 2016).

4.5.4. UR 4.4 – risk and uncertainties

CR 4.4.1 & CR 4.4.2 – risk informed approach and uncertainties. Risk informed approach has not yet been applied to MSRs. But a risk informed methodology has been suggested by Mandell et al., at Ohio state university for molten salt reactors (Mandelli et al., 2007). The uncertainties associated with MSRs due to lack of experimental knowledge have not been dealt with so far. But the innovative designs of MSRs show enough promise to warrant interest from all over the world. A comprehensive PSA needs to be done to investigate all the uncertainties.

5. Discussion

In the area of safety of INPRO methodology, there are 4 basic principles under which there are 14 user requirements. These user requirements further contain 37 criteria, which are supplemented with indicators against which an assessment can be made. Each indicator has its own acceptance limits. If this acceptance limit is met, then the INS is safe against that requirement. In the present research, these 37 criteria have been applied to the concept of MSR and the assessment has been

made based on the data available in open literature. After performing the assessment, it is observed that the molten salt technology ticks most of the boxes in terms of safety. Most of the criteria specified by INPRO methodology are met, but there are some that could not be met based on the data available. For example, CR3.1.1 and CR 3.2.1 which are concerned with the occupational and public doses. Although the operating conditions and inherent features of MSRs promise a lower release, yet no conclusive figures are given by current designers. Then there are some criteria that could not be met by current MSRs designs. e.g. CR 4.1.1, definition of safety concept. The safety concept is not defined because of the absence of detailed uncertainty-free models and lack of proof of concept. Similarly CR 4.2.2, which is related with the availability of computer codes is not met. Complete computer codes for MSRs have not been developed so far, hence this criterion could not be met. The same goes for CR 4.4.1, which relates to risk-informed approaches being used to analyze the MSRs. These criteria are not met but they are not a knock on the promise that MSRs show because this technology has not been fully developed. It just shows the power of INPRO methodology since this methodology helps the assessor in recognizing the areas that need further research and improvement.

Despite the unavailability of detailed experimental information, most of the requirements set by INPRO have been met for MSR concept. There is one peculiar criterion – CR 1.3.4, which concerns the barriers against release. Since the regulations that are in order today are for solid fuel reactors with high operating pressures, barriers are designed to withstand against such high pressures in order to meet those regulations. In MSRs, there is molten core and the pressures involved are quite low. Hence the type of barriers required for MSRs are different than those in conventional solid fuel reactors. Hence, new set of definitions should be introduced to cover the concept of barriers against release of radioactivity in molten salt reactors. INPRO methodology is based on the fundamental safety objective of protection against harmful radiations. The release of such harmful radiations is not related to melting of core in case of MSRs.

Currently at this stage of development, INPRO methodology cannot be applied completely to MSR. But the application of INPRO certainly points out the issues or deficiencies in the area of safety of MSR concept and hence direct attention to those areas as explained above. Once the concept is matured, INPRO methodology may be applied again to assess the safety of the concept.

6. Conclusion

INPRO methodology is a very powerful tool as it applies to both operational reactors and reactors in the design phase. In this research, safety requirements of INPRO methodology are applied to the concept of molten salt reactors. MSRs are judged based on the data available in open literature. Most of the requirements are satisfied by current concepts of MSR. Few requirements that are not completely satisfied by molten salt reactors help in directing the attention towards those specific areas. This research further signifies the fact that MSRs are inherently very safe reactors. In terms of robustness of design; safety margins; introduction of passive safety systems; and automation in modern reactors; MSR is indeed a good option. The technological challenges posed by the novelty of MSRs demand further research. The shortcomings in MSRs are not inherent in their concept but are due to the limited research completed on them. As per INPRO methodology, further research should be conducted to make molten salt reactors a tangible reality.

The application of INPRO method of assessment is a good test of INPRO set of requirements as well. Since the concept of molten salt reactor (MSR) is quite different from currently available designs of solid fuel reactors, the combination of INPRO requirements and MSR concept helped in realizing both the strong areas and the gaps that need further improvement.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pnucene.2019.103099.

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